Consumer Choice in Differentiated-Product Markets: The BLP Model through a Systems and Control Lens

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1 Scope

Ariel Pakes' central contribution to the modelling of consumer choice is the random-coefficients discrete-choice demand system of Berry, Levinsohn and Pakes (BLP) [2]. BLP recovers heterogeneous preferences and firms' markups from aggregate market data while accounting for the fact that prices respond to demand conditions that the analyst does not observe. This note reviews the model and the literature around it, states it mathematically, and reads it from the perspective of systems and control theory, with the aim of identifying where game theory can enter the research of a control group.

2 Literature

Foundations. The random-utility multinomial logit of McFadden [12] models a consumer choosing the alternative with the highest utility under i.i.d. extreme-value noise. Its closed-form choice probabilities come at the price of the independence of irrelevant alternatives (IIA): substitution patterns depend only on market shares. Berry [4] showed that with market-level data the share equations can be *inverted* to recover a mean-utility index in which the unobserved product quality enters linearly, turning a non-linear endogeneity problem into a linear instrumental-variables (IV) problem.

The BLP model. BLP [2] combined the inversion with random coefficients on product characteristics, which breaks IIA, and with a Bertrand–Nash pricing game on the supply side, estimating both jointly by GMM on U.S. automobile data. Nevo [13] used the framework to measure market power in the cereal industry and became a standard reference for its implementation, while BLP [3] added consumer-level "micro moments" to sharpen identification of heterogeneity.

Identification and computation. Berry and Haile [5] established nonparametric identification of demand from market-level data under an index restriction and suitable instruments. On the numerical side, Dubé, Fox and Su [7] showed that loose tolerances in the inner fixed-point loop bias estimates and proposed a constrained-optimisation (MPEC) formulation; Conlon and Gortmaker [6] consolidated best practices (instruments, integration, optimisation) in the PyBLP package.

Dynamics and Pakes' broader programme. Ericson and Pakes [8] model industry evolution as a stochastic game with entry, exit and investment, solved for a Markov-perfect equilibrium; this is the dynamic supply-side counterpart to static BLP demand. Gowrisankaran and Rysman [9] make demand itself dynamic, with forward-looking consumers who time durable-good purchases. Pakes, Porter, Ho and Ishii [14] replace point-identifying equilibrium conditions with *moment inequalities* derived from revealed preference, yielding set estimates robust to agents' imperfect information.

3 Mathematical Model

Consider markets $t = 1, \dots, T$, products $j = 1, \dots, J_t$ and an outside good $j = 0$. Consumer i obtains utility

$$u_{ijt} = \underbrace{x_{jt}^{\top}\beta - \alpha p_{jt} + \xi_{jt}}_{\delta_{jt}} + \underbrace{\textstyle\sum_k x_{jtk}\,\sigma_k \nu_{ik}}_{\mu_{ijt}(\theta_2)} + \varepsilon_{ijt}, \qquad u_{i0t} = \varepsilon_{i0t}, \tag{1}$$

with observed characteristics x_{jt}, price p_{jt}, unobserved quality ξ_{jt}, taste shocks $\nu_i \sim F$ and ε_{ijt} i.i.d. type-I extreme value. Integrating over consumers gives the market shares

$$s_{jt}(\delta_t; \theta_2) = \int \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_{k=1}^{J_t} \exp(\delta_{kt} + \mu_{ikt})}\, dF(\nu_i). \tag{2}$$

Inversion. For given θ_2, the observed shares $\mathcal{S}_t$ are matched by the unique fixed point of

$$\delta_t^{h+1} = \Phi(\delta_t^h) = \delta_t^h + \ln \mathcal{S}_t - \ln s_t(\delta_t^h; \theta_2), \tag{3}$$

which is a contraction [2]. *Estimation.* Since prices are correlated with ξ, one recovers $\xi_{jt}(\theta) = \delta_{jt}(\theta_2) - x_{jt}^\top \beta + \alpha p_{jt}$ and solves $\min_\theta \bar{g}(\theta)^\top W \bar{g}(\theta)$ with $\bar{g}(\theta) = \frac{1}{N}\sum_{jt} Z_{jt}\,\xi_{jt}(\theta)$, where the instruments Z (cost shifters, rivals' characteristics) satisfy $\mathbb{E}[\xi_{jt} \mid Z_{jt}] = 0$. *Supply.* Firm f owning products $\mathcal{F}_f$ maximises $\sum_{j\in\mathcal{F}_f}(p_j - mc_j)\,s_j(p)$. The Bertrand–Nash first-order conditions give

$$p = mc + \Delta(p)^{-1} s(p), \qquad \Delta_{jk} = -\frac{\partial s_k}{\partial p_j}\,\mathbf{1}\{j,k \text{ same owner}\}, \tag{4}$$

so observed prices and estimated demand reveal marginal costs and markups.

4 A Systems and Control Reading

- **Price endogeneity is closed-loop identification.** Firms set the input p through a feedback law, (4), that reacts to the disturbance ξ. As in closed-loop system identification [11], regressing output on input is biased; cost shifters play the role of an external reference signal, and BLP instruments are the econometric analogue of IV methods in system identification.
- **Logit choice is entropy-regularised optimisation.** The logit share vector solves $\max_{s\in\Delta}\langle s, u\rangle + H(s)$ with $H(s) = -\sum_j s_j \ln s_j$, i.e. a softmax (smoothed best response). Mixed logit is a heterogeneous population of such agents.
- **Inversion and equilibrium are fixed-point problems.** (3) is a contraction iteration whose modulus determines convergence speed; the inner tolerance issue of [7] is an error-propagation question, and MPEC turns the nested loop into a single constrained programme, as in MPC.
- **Recovering costs is an inverse game.** (4) inverts Nash conditions to recover agents' objectives from observed equilibrium play, the game-theoretic counterpart of inverse optimal control.
- **Shares are a population state.** With shares as a state on the simplex, the logit dynamics $\dot{s} = \sigma(u(s)) - s$ of population games [15] give equilibrium as a rest point whose stability can be analysed with Lyapunov and passivity tools; with a continuum of heterogeneous, forward-looking consumers the natural object is a mean-field game [10].
- **Dynamic models are stochastic dynamic games.** Ericson–Pakes equilibria are feedback (Markov) Nash equilibria in the sense of Başar–Olsder [1], and moment inequalities [14] parallel set-membership estimation in robust control.

5 Integrating Game Theory: Research Directions

1. **Feedback Nash pricing with demand dynamics:** couple a dynamic demand state (adoption, inventories, network effects) with firms' pricing as a differential or difference game; study existence, uniqueness and stability of feedback equilibria [1, 8].
2. **Learning in games and adaptive control:** firms that estimate BLP demand online while pricing face a dual-control trade-off between exploration and exploitation; convergence of best-response or gradient learning to the Bertrand–Nash equilibrium is an open systems question.
3. **Leader–follower design:** a regulator or platform (Stackelberg leader) steers a population of logit consumers and competing firms through taxes, subsidies or information, linking incentive design with population-game stability [15].
4. **Identification of games from data:** closed-loop identification theory [11] can inform instrument design, excitation conditions and inverse-game estimation of agents' objectives in multi-agent systems.
5. **Large-population limits:** mean-field games [10] offer a tractable model of many strategic consumers with BLP-style heterogeneity.

In summary, BLP is at its core an estimation problem for a static Nash equilibrium of a population game under feedback; making the dynamics, learning and control explicit is where control theory can contribute most.

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